Unit Converter

Project notes · 6 November 2026

What it is

A single-page converter app with a home menu and three separate screens: length, weight and temperature. Everything runs in the browser, so no data leaves the page and no account is needed.

Screens and navigation

Screen	Address	What it does
Menu	#home	Three cards, each showing its units and a sample conversion. Opens by default.
Length	#length	Converts between 9 length units. Base unit is the metre.
Weight	#weight	Converts between 7 mass units. Base unit is the kilogram.
Temperature	#temperature	Converts between Celsius, Fahrenheit and Kelvin, with a gauge.

Navigation uses the address hash, so each screen has its own link and the browser back button works. Each screen has a back link and a tab bar to jump between screens. Values you enter are kept when you move between screens.

How a converter screen works

- Type a value under **From**, choose the from and to units, and the result updates as you type.
- **Swap** flips the two units and moves the result into the input.
- **Copy result** copies the plain number to the clipboard.
- **Every unit at once** lists the value in all units. Clicking a row sets it as the target unit.
- **Try a real one** loads a reference value, such as a marathon or a litre of water.
- Invalid input shows a short message. The temperature screen warns below absolute zero.

Units and factors

Length (metres per unit)

Unit	Metres	Unit	Metres
Millimetre (mm)	0.001	Foot (ft)	0.3048
Centimetre (cm)	0.01	Yard (yd)	0.9144
Metre (m)	1	Mile (mi)	1609.344
Kilometre (km)	1000	Nautical mile (nmi)	1852
Inch (in)	0.0254		

Weight (kilograms per unit)

Unit	Kilograms	Unit	Kilograms
Milligram (mg)	0.000001	Ounce (oz)	0.028349523125
Gram (g)	0.001	Pound (lb)	0.45359237
Kilogram (kg)	1	Stone (st)	6.35029318
Tonne (t)	1000		

Every factor above is exact by definition. Ounces and pounds are avoirdupois, and 1 stone is 14 pounds.

Temperature formulas

From	To	Formula
Celsius	Fahrenheit	°F = °C x 9/5 + 32
Fahrenheit	Celsius	°C = (°F - 32) x 5/9
Celsius	Kelvin	K = °C + 273.15
Kelvin	Celsius	°C = K - 273.15
Fahrenheit	Kelvin	K = (°F - 32) x 5/9 + 273.15
Kelvin	Fahrenheit	°F = (K - 273.15) x 9/5 + 32

Absolute zero is 0 K, -273.15 °C or -459.67 °F. Reference points on the gauge: water freezes at 0 °C, the body sits near 37 °C, and water boils at 100 °C.

Design and build notes

- Each screen has its own accent: blue for length, amber for weight, vermilion for temperature.
- Light and dark themes follow the viewer's system setting.
- Type: Familjen Grotesk for headings, IBM Plex Sans for text, IBM Plex Mono for numbers.
- Layout stacks to one column at phone width and nothing scrolls sideways.
- Conversions go through a base unit (metres, kilograms, Celsius), so any pair works with one formula per unit.
- Numbers are rounded to 10 significant digits to hide floating-point noise, for example 69.85322498 kg.
- Large or tiny results switch to exponent form beyond 1e15 and below 1e-6.

Worked examples

Input	Result
26.2 mi	42.1648128 km
154 lb	69.85322498 kg
98.6 °F	37 °C

Input	Result
0 K	-273.15 °C

Possible next steps

- More units, such as area, volume, speed or data size, each as a new menu card.
- Remember the last units chosen on each screen between visits.
- A history list of recent conversions with one-tap reuse.